Total No. of Questions : 12]

SEAT No. :

P3245

[Total No. of Pages : 2

[5353] - 108
T.E. (Civil) (Semester - II)
FOUNDATION ENGINEERING
(2012 Pattern)

Time :2½ hours] ***[Max. Marks :70***

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q. 3or Q. 4,Q.5 or Q.6 Q.7 or Q.8, Q.9 or Q.10. and, Q.11 or Q.12.***
2) ***Neat diagrams must be drawn whenever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data ,if necessary and mention it clearly.***

Q1) Explain with neat sketch Seismic Refraction method of Soil Exploration. **[7]**

OR

Q2) Define Disturbed Sample and Undisturbed Sample. State the factors affecting on Sample Disturbance **[7]**

Q3) Differentiate between Local shear failure and General shear failure. **[6]**

OR

Q4) Differentiate between Terzaghi's Bearing Capacity Theory and Meyerhof's Bearing Capacity Theory. **[6]**

Q5) Define i) Immediate Settlement ii) Consolidation Settlement iii) Secondary Settlement with variation of settlement with time. **[7]**

OR

Q6) Explain with neat sketch laboratory one dimensional consolidation test **[7]**

Q7) a) Write short note on i) End Bearing Pile. ii) Friction Pile **[6]**

b) Explain with neat sketch various component of well foundation. **[6]**

c) Discuss the necessity of pile foundation. **[4]**

OR

Q8) a) Define Negative Skin Friction and state the situation where Negative Skin Friction can be anticipated. **[6]**

b) Draw the neat sketch with application i) Open Caisson ii) Box Caisson iii) Pneumatic Caisson. **[6]**

c) Write a short note on caisson disease. **[4]**

Q9) a) Write short note on i) Earth fill coffer dam. ii) Rockfill Cofferdam. **[6]**

b) List out the various techniques of Ground soil improvement. Explain any one. **[6]**

c) Explain any four engineering problem associated with black cotton soil. **[4]**

OR

Q10) a) Write short note on i) Bored Pile ii) Under reamed Pile. **[6]**

b) Plot the Pressure Distribution diagram for i) Cantilever Sheet Pile ii) Anchor Sheet Pile. **[6]**

c) State the characteristic of Black Cotton soil. **[4]**

Q11) a) State the various function of geosynthetic materials and explain any two function with suitable example **[7]**

b) What is liquefaction? Discuss the effect of liquefaction. **[6]**

c) Write a short note on classification of Geosynthetic material. Explain any one. **[5]**

OR

Q12) a) Write down the application of geosynthetic material in different area with suitable example with sketch. **[7]**

b) Define the term i) Earthquake ii) Epicenter iii) Focus iv) Magnitude. **[6]**

c) Differentiate between P-wave and S-wave. **[5]**

❀ ❀ ❀